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Palaeomagnetism and Magnetic Polarity Zones in the Jurassic Dufek Intrusion, Pensacola Mountains, Antarctica*

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Summary

The Dufek intrusion, of the Pensacola Mountains, Antarctica, is a stratiform mafic complex consisting of layered gabbro with subordinate amounts of anorthosite, pyroxenite and granophyre, forming a little-deformed sheetlike intrusion at least 6 km thick. Its age is estimated as Jurassic. Of 183 oriented hand samples originally collected from the intrusion for palaeomagnetic studies, 98 proved to be magnetically stable, including 61 samples of normal polarity, 30 of reversed polarity, and seven that are intermediate. The distribution of polarities within the intrusion makes it impossible to rule out either self-reversal or geomagnetic field reversal as the cause of the dual magnetic polarities; possibly both processes were operative. Both intensity of remanent magnetization and magnetic susceptibility generally increase with stratigraphic height in the intrusion. Mean directions of remanent magnetization for the two polarity groups are very nearly antiparallel, and when combined yield a south palaeomagnetic pole at 56·5° S., 168° W., for the Dufek intrusion. This pole lies near poles previously reported for the Jurassic Ferrar dolerite. Available data are not sufficient to define the behaviour of the pole in relation to Antarctica during most of the Mesozoic. Palaeomagnetic evidence tends to support the proposition that West and East Antarctica were independent entities in pre-Tertiary time, and to suggest that there has been significent Tertiary movement of crustal blocks within West Antarctica.

1. Introduction and geological summary

This report describes the results of studies of the remanent magnetism of layered mafic rocks from the Dufek intrusion, Pensacola Mountains, Antarctica

Among interesting recent applications of palaeomagnetic data are attempts by several authors (McElhinny 1971; McElhinny *et al.* 1968; Creer 1967, 1970; Zijderveld 1968) to reconstruct Gondwanaland in its pre-drift configuration. The question of the proper reconstruction has bearing upon such diverse topics as the cause of Palaeozoic glaciation (Crowell & Frakes 1970), the history and tectonics of West Antarctica (Hamilton 1967; Schopf 1969), and the possibility of the existence of recurrent episodes of continental drift (Irving 1966; Briden 1967). A weak link in all proposed palaeomagnetic reconstructions is the apparent polar wandering curve for Antarctica, which is based on too few data to be of much assistance in palaeomagnetic reconstructions. For instance, in his recent review Creer (1970) presents a polar wandering curve for Antarctica consisting of two line segments connecting a total of three reliable positions of the palaeomagnetic pole.

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The geology of the Dufek intrusion was first examined during the IGY Ellsworth Station traverse in December 1957 (Aughenbaugh 1961; Walker 1961). Detailed geological field studies were made during 1965–66 by a U.S. Geological Survey expedition. The geology of the intrusion has been described in recent papers by Ford & Boyd (1968, 1969) and Ford (1970). According to these authors the Dufek intrusion is a layered sequence of gabbro and related rock, forming a roughly sheetlike body estimated to be at least 6 km thick. Aeromagnetic evidence indicates that the intrusion underlies an area of at least 9500 km^2 (Behrendt, Meister & Henderson 1966), much of which is covered by ice. Extensive exposures occur in two areas. A thickness of approximately 1·9 km of the stratigraphically lower part of the intrusion, consisting of layered gabbro, anorthosite and pyroxenite, is exposed in the Dufek Massif (Fig. 1). An ice-filled valley about 60 km wide separates the Dufek Massif rocks from exposures of the upper part of the intrusion, in the Forrestal Range. Forrestal Range rocks have an exposed thickness of approximately 1·6 km, consisting chiefly of a layered gabbro that is more iron rich and more oxidized that the layered gabbro of the Dufek Massif. Granophyre, 0·3 km thick, caps the Forrestal Range exposures (Ford 1970). Layering is present throughout most of the intrusion, and in places the rock fabric shows evidence that suggests transport of mineral grains in a

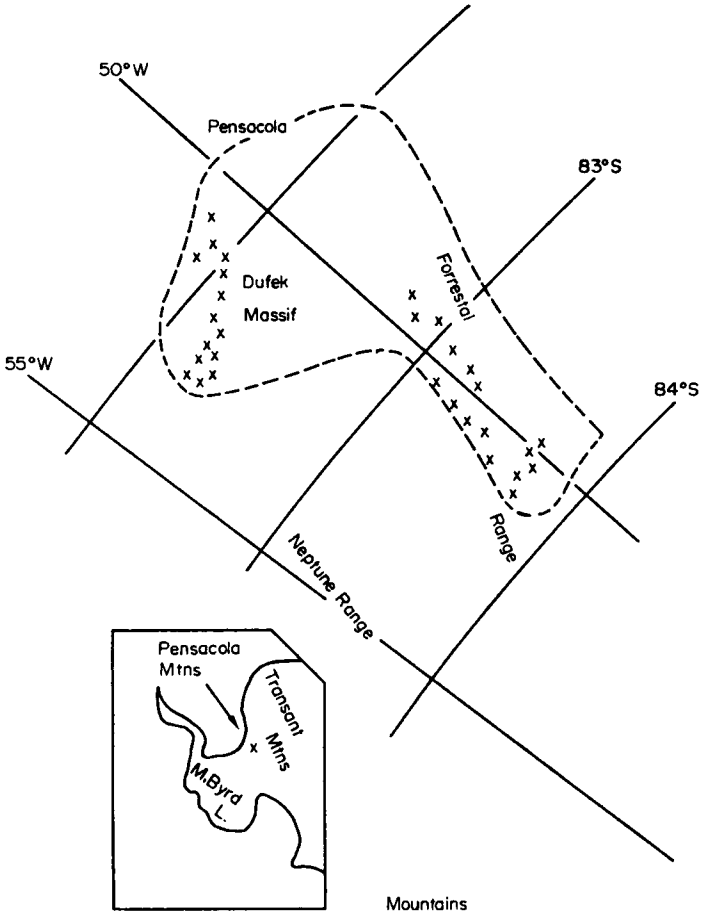


FIG. 1. Index map showing the location of the Dufek intrusion. Modified from Ford (1967).

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convecting fluid medium. Slight dips measured in layered exposures indicate that the intrusion has been subjected to postconsolidation warping (Ford 1971). K–Ar plagioclase dates by R. W. Kistler (quoted by Ford 1971) cluster about 168 ± 5 My, indicating that the Dufek intrusion is Middle Jurassic in age.

Rock magnetic data previously reported for the Dufek intrusion consist of a preliminary account of the palaeomagnetism by Beck, Ford & Boyd (1968) and a brief analysis of magnetic intensities by Beck & Griffin (1971). A tabulation of magnetic intensities was given by Griffin (1969). The earlier palaeomagnetic report was based on NRM directions of 43 magnetically stable samples from the Dufek Massif section. The results reported here supersede the earlier palaeomagnetic report.

2. Methods of investigation

Samples for this investigation were collected during 1965–66 by members of a helicopter-supported field party of U.S. Geological Survey geologists. A total of 183 oriented hand samples was collected by A. B. Ford, W. W. Boyd, W. H. Nelson and D. L. Schmidt. Samples were collected during measurement of sections through the entire exposed stratigraphic thickness of the intrusion, and were spaced throughout most of its exposed area. On the assumption that the Dufek intrusion is a gabbroic sheet 6 km in thickness, Fig. 2 of Jaeger (1957) suggests that the time required for the Curie temperature isothermal surface (assumed 500 °C) to pass from the margins of the sheet to its centre was on the order of 0·25 My. If the isothermal surface moved at a constant velocity, a stratigraphic thickness of about 0·5 m of gabbro anywhere within the sheet should have become magnetized during a period of about 50 years. In this simplified model the isothermal surfaces are assumed to be horizontal, and therefore are roughly parallel to the igneous layering. On the arbitrary assumption that samples which are magnetized less than 50 years apart do not provide independent estimates of the direction of the geomagnetic field, samples separated by a stratigraphic interval of 0·5 m or less were averaged. This averaging procedure yielded 61 independent sampling localities in about 1·5 km of Forrestal Range gabbro and 79 independent localities in about 1·8 km of gabbro from the Dufek Massif.

The accuracy of the orientation technique used is difficult to assess, but probably is relatively low and variable from sample to sample. Samples were collected by four geologists working independently, all using the 'strike-and-dip' technique to orient natural planar surfaces. One source of potential error arises from the use of the magnetic compass for orientation purposes, since it was not possible under the conditions of the field work to correct for minor declination anomalies that might have been caused by the rock itself. Local declination anomalies are likely to have been particularly strong in the Forrestal Range gabbros, some of which have high intensities of magnetization. Thus, individual orientations are likely to contain errors of 4°–5°, on the average, and the strongly divergent directions of some of the stable samples, termed 'intermediate' in subsequent sections, may be the result of large orientation errors. Orientation errors thus contribute to the dispersion of directions of remanent magnetization about their mean but, being random, are unlikely to have significantly altered the direction of the mean.

A single laboratory specimen was prepared from each oriented hand sample and the remainder of the rock retained for chemical and petrographic studies. Directions of remanent magnetization were measures on a duplicate of the U.S. Geological Survey spinner magnetometer described by Doell & Cox (1965). Certain very weakly magnetized samples were measured on a more sensitive spinner magnetometer designed by Phillips & Kuckes (1967). Magnetic susceptibilities were measured in low magnetic fields by an inductance bridge (Anderson 1961) and checked on a commercial bridge of similar design. All specimens were subjected to progressive cleaning in alternating magnetic fields of 75, 150 and 300 Oe peak value, using a 3-axis tumbler to randomize

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directions within the specimen with respect to the axis of the coil. Selected specimens also were cleaned at 35·5 and 600 Oe. As expected from the great diversity in lithology and chemical composition found in the Dufek intrusion, magnetic stability also varied greatly. This is illustrated in Fig. 2, which shows composite demagnetization curves (modified from Doell 1969) for stable and unstable specimens of normal magnetic polarity.

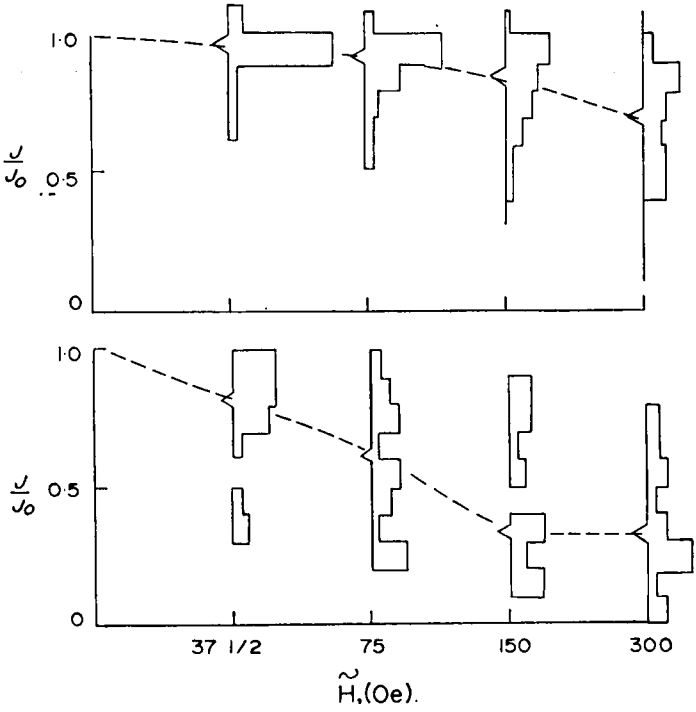


FIG. 2. Composite demagnetization curves for *N* polarity samples. Upper diagram represents stable samples, lower diagram unstable samples. $\tilde{H}$, peak value of alternating magnetic field; J, intensity of remanent magnetization at the indicated demagnetization level; J_0, intensity of natural remanent magnetization.

Fig. 2 suggests that certain samples are highly unstable. In order to avoid obscuring the record of the Jurassic geomagnetic field preserved in the primary remanent magnetization of Dufek intrusion rocks, it was necessary to establish criteria for the rejection of unstable specimens. The method chosen was the 'stable end point' technique of McElhinny & Gough (1963), wherein a specimen is judged to be free of secondary magnetization if its direction of remanent magnetization ceases to change with increase in the demagnetizing field. For the purposes of the present study a stable end point was reached if, for three successive demagnetization levels, the direction of remanent magnetization changed by less than 5°. Alternating field magnetic cleaning was not pursued systematically beyond 600 Oe because the direction of many specimens began to change erratically at or above that point. Apparently imperfections in the demagnetization process added spurious magnetic moments to some specimens. Spurious moments probably of similar origin have been analysed by Doell & Cox (1967). The problem of imperfections in the demagnetization process was particularly acute in this investigation because, as shown by their response to a.f. demagnetization, many specimens have relatively low spectra of microscopic coercivity.

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3. Magnetic polarity

Fig. 3 shows that directions of remanent magnetization of stable samples from the Dufek intrusion, after magnetic cleaning and tectonic correction, tend to cluster into two nearly antiparallel groups—a normal (N) polarity group in the first quadrant of the upper hemisphere, and a reverse (R) group in the third quadrant of the lower hemisphere. More than 90 per cent of the magnetically stable samples can be associated unambiguously with either the *N* or the *R* group. A few are clearly intermediate (*I*), however, and must be eliminated before calculation of average directions of magnetization. For the purposes of this study samples were classified as intermediate if they diverged from the mean by more than 55°, this being about twice the angular standard deviation. The origin of stable intermediate directions is discussed below.

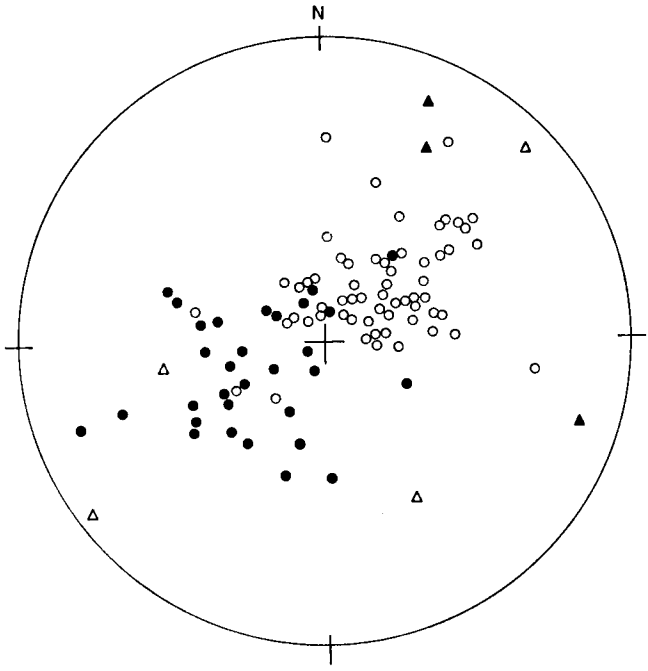


FIG. 3. Equal-area plot of directions of remanent magnetization of stable samples from the Dufek intrusion, after alternating field demagnetization and tectonic correction. Open symbols, upper hemisphere; filled symbols, lower hemisphere. Triangles show samples with direction intermediate between the normal and reverse groups.

Table 1

Mean directions of remanent magnetization for normal and reverse groups of stable samples, after magnetic cleaning and tectonic correction

Group	*n*	$\bar{D}$	$\bar{I}$	*k*	α_{95}	δ
N	60	45·5°	−69·0°	14·2	5·0°	21·5°
R	31	247·5°	68·0°	10·3	8·5°	25·0°

Explanation: *N*, normal group; *R*, reverse group; *n*, number of samples; $\bar{D}$, mean declination; $\bar{I}$, inclination; *k*, estimate of precision parameter (Fisher 1953); α_{95}, radius of circle of 95 per cent confidence; δ, angular standard deviation (Wilson 1599). Samples from the Dufek Massif have been corrected for an average dip of 7° SE; samples from the Forrestal Range have been corrected for an average dip of 10° SE.

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Table 1 shows average directions for the *N* and *R* groups, after tectonic correction. The means for the two groups are 172° from one another, and at the 95 per cent probability level the two groups are exactly antiparallel.

The stratigraphic distribution of polarity in the Dufek intrusion is shown in Fig. 4. Both the Dufek Massif and the Forrestal Range sections divide naturally into two zones, a zone of uniform polarity and a zone of mixed polarity. Approximately the top 850 m of exposures in the Dufek Massif furnish magnetically stable samples that are exclusively of *N* polarity (but include two *I* samples). Roughly the bottom 600 m of exposures in the Forrestal Range are of uniform *R* polarity. In contrast, the upper-

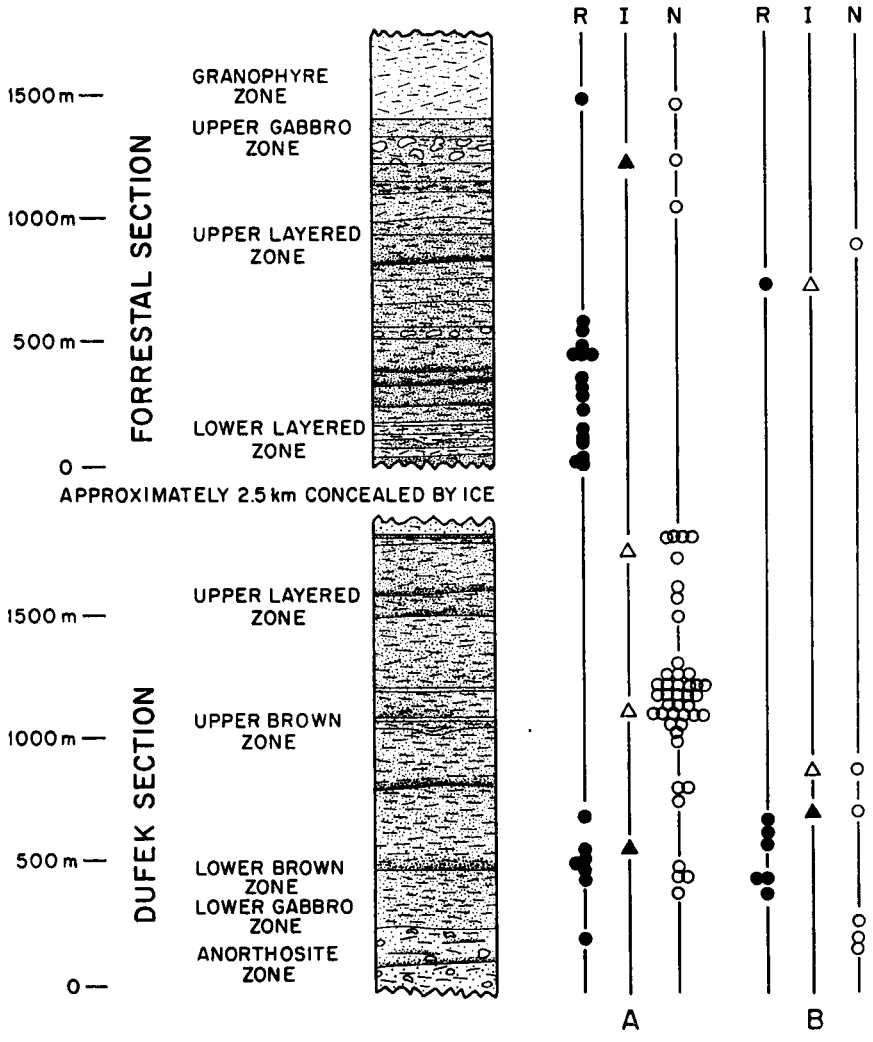


FIG. 4. Stratigraphic distribution of magnetic polarities of magnetically stable samples from the Dufek intrusion. *R*, *I*, *N*, indicate reverse, intermediate, and normal, respectively. For intermediate samples, filled triangle indicates the sample direction is nearer reverse than normal polarity; open triangle indicates the opposite. Stratigraphic location of samples in column A is known with considerable certainty; position of samples in column B is inferred from indirect geological evidence and may be slightly in error. Eight samples are omitted altogether because their location is too poorly known. For further details of magnetic stratigraphy in the Dufek intrusion see Griffin (1969). Lithological column modified from Ford & Boyd (1968).

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most Forrestal Range exposures and the lowermost Dufek Massif exposures show an abrupt alternation between *N* and *R*. These zones also contain five of the seven stable *I* samples. A significant observation not brought out by Fig. 4 is that the apparent alternation in polarity in the mixed zones is not observed in individual sampling traverses. For instance, the Dufek mixed zone is defined on the basis of mixed polarity in samples collected throughout the lowermost 900 m of exposures in the intrusion. However, these samples were collected in six different sampling traverses spaced over a horizontal distance of about 17 km. Each traverse is consistent within itself, showing (except for two *I* samples) either samples of a single polarity or samples of one polarity below and the opposite polarity above.

Mixed polarities are found in many palaeomagnetic studies. For instance, in many deep-sea cores an alternation of discrete zones of magnetic polarity has been observed (Hays & Opdyke, 1967; Watkins 1968) and generally attributed to reversal of polarity of the geomagnetic field during the slow accumulation of deep oceanic sediments. A similar phenomenon has been reported by many authors (e.g. Watkins 1965) for thick sequences of lava extruded over periods of the order of 10^4 years or more. So successful has been the hypothesis of geomagnetic reversal in explaining 180° alternations of magnetic directions in rocks that most investigators look first to field reversals when faced with an instance of mixed polarity. However, it also has been established that some rocks have the property of acquiring remanent magnetism antiparallel to the direction of the ambient magnetic field (Uyeda 1958). Instances of self-reversal apparently require somewhat unusual compositions and/or geometrical arrangements of magnetic minerals (Irving 1964, pp. 159–168); thus self-reversal seems to be rare in rocks normally used for palaeomagnetic studies. Nevertheless, in certain cases it has been necessary to invoke self reversal to explain differences in magnetic polarity, as when samples of opposite magnetic direction occur within the same outcrop (Merrill & Grommé 1969). Merrill & Grommé argue that self-reversal in rocks whose magnetic behaviour is dominated by hemoilmenite may be more common than usually supposed. However, field evidence indicates that self-reversal is not common in gabbroic rocks. For example, of 165 stable sites representing the full range of lithology of strongly differentiated Keeweenawan gabbroic rocks in Minnesota (Beck 1970), there are only two instance of inconsistent polarity that might be attributed to self-reversal. In strongly differentiated diabase from south-eastern Pennsylvania, (Beck 1965 and unpublished data), 105 sites all have consistent polarity. Other investigators have reported similar results. On the basis of field evidence it seems safe to conclude that self-reversal of remanent magnetization is extremely rare in gabbroic rocks.

The cause of alternating magnetic polarities in the Dufek intrusion is uncertain. Ideally, zones of alternating magnetic polarity caused by reversal of the geomagnetic field during cooling of a horizontal sheet would show a simple, roughly symmetrical pattern. Fig. 5 shows a hypothetical distribution of polarity zones in an intrusive sheet which began to acquire its remanent magnetization in a geomagnetic field of N polarity and experienced three changes of geomagnetic polarity during the magnetization process. The horizontal plane of symmetry shown in Fig. 5 is displaced upward from the centre of the sheet to reflect the temperature distrubution likely to result from crystal settling during crystallization and differentiation of a sheet of gabbroic magma. Clark (1969, Fig. 8) infers a similar pattern for the much thinner Marys Peak gabbroic sill in western Oregon. The exact amount of upward displacement is uncertain. Details of the pattern will vary with such factors as original temperature distribution, shape of the sill, distribution within the sill of blocking temperatures (temperature of acquisition of thermoremanent magnetization), and temperature gradient and thermal conductivity of the country rock. On the basis of existing information, Fig. 5 could account for the distribution of magnetic polarities in the Dufek intrusion without need to invoke self reversal, provided that the blocking-

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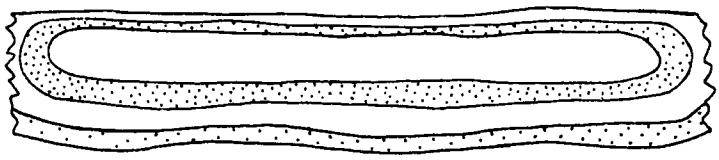


FIG. 5. Hypothetical polarity zones developed in a sheetlike intrusion by cooling through three polarity transitions. Stippled pattern, *N* polarity; blank pattern, *R* polarity.

temperature isothermal surfaces were somewhat irregular when they first entered the intrusion. Initial irregularity of this sort would be expected unless the country rock was entirely homogeneous with respect to thermal properties and the shape of the intrusion was entirely regular.

The distribution of magnetic polarities can also be explained by the self-reversal mechanism. Self-reversal is a mineralogical property and thus should correlate with rock type rather than with time of magnetization. In many layered-mafic intrusions such things as gross chemical composition, composition of specific minerals, relative proportion of specific minerals, and ferric–ferrous ratio are observed to vary systematically with height above base. This is certainly true of the Dufek intrusion (Ford & Boyd 1968; Ford 1970). If oxide mineralogy changes systematically with stratigraphic position, zones of rock having self-reversing properties might develop parallel to the igneous stratigraphy. For instance, the Dufek *N* zone (Fig. 4) could represent self-reversed material and the Forrestal *R* zones could record the true polarity of the geomagnetic field, or vice versa. The two mixed zones by this model might reflect sections in the intrusion where relatively rapid changed in iron titanium oxide mineralogy produced an alternation between self-reversing and ordinary gabbroic rock. An explanation based on self-reversal alone is unlikely, in view of the extreme rarity of recorded examples of self-reversal in gabbroic rocks. Self-reversal may have modified the pattern set by reversal of the geomagnetic field, however. Self-reversing properties in Dufek rocks will be investigated during projected studies of the mineralogy and magnetic properties of the oxide minerals.

The explanation for the existence of stable samples with intermediate (I) directions of remanent magnetization also is uncertain. Intermediate samples theoretically could represent: (1) magnetization while the geomagnetic field was changing polarity; (2) magnetization during an aborted polarity transition (Watkins 1969); (3) magnetization during brief excursions of the main dipole field; (4) two ' hard ' components of magnetization, perhaps a chemical remanent magnetization superimposed on a primary thermoremanent magnetization; (5) a very hard isothermal remanent magnetization, as might be produced by a lightning discharge in close proximity to the sampling locality, (Cox 1161; Graham 1961); (6) failure to properly orient the hand samples in the field. In addition, several *I* samples are just outside the arbitrary limits set for the *N* and *R* groups and may merely represent extreme scatter. Cause (1) is unlikely because the *I* samples are not preferentially located near boundaries between zones of opposite polarity. Causes (1) and (2) are unlikely in view of the fact that *I* samples have unremarkable remanent magnetic intensities (Fig. 6) and therefore probably were magnetized in magnetic fields of ordinary strength. Cause (5) is virtually eliminated by this same observation. As observed above, cause (6) is a likely contributor to at least some of the *I* directions.

4. Magnetic intensities

NRM (natural remanent magnetization) intensities and magnetic susceptibilities in the Dufek intrusion show the usual increase with height above base (Jaeger &

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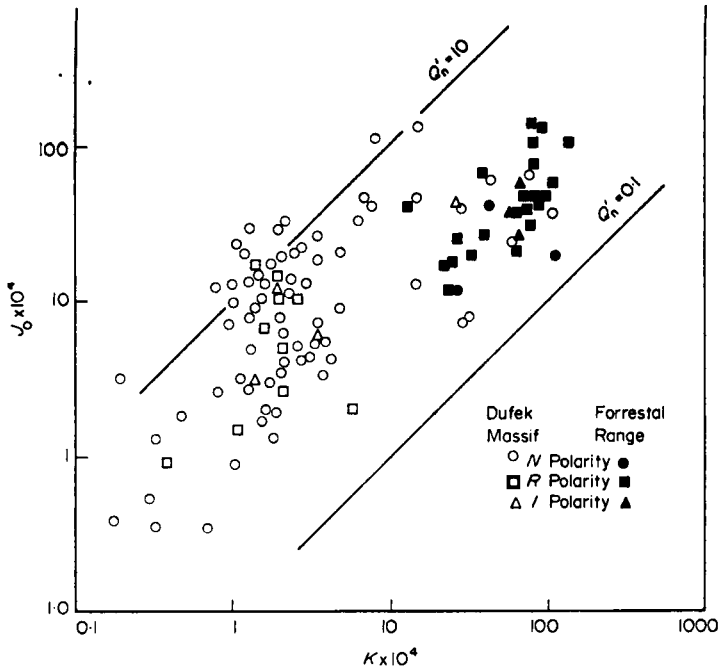


FIG. 6. Intensities of natural remanent magnetization and magnetic susceptibilities of stable samples from the Dufek intrusion. J_0, natural remanent magnetization, in emu cm^{-3}; k, volumetric susceptibility, in emu/Oe. $Qn' = J_0/K$.

Joplin 1955; Beck 1966; 1970). This increase is reflected in Fig 6 by the higher intensities and susceptibilities shown by magnetically stable samples from the Forrestal Range. There is no apparent correlation between NRM intensity or magnetic susceptibility and magnetic polarity. *R* samples (Fig. 6) appear to have greater than average intensities and susceptibilities, but this is due simply to the fact that most *R* samples were obtained near the top of the intrusion, where relatively high magnetizations are expected. Q_n, the ratio of NRM intensity (J_0) to magnetic susceptibility (K) (Irving 1964, p. 11) varies in stable samples from just under 1 to just over 10. For any reasonable value of geomagnetic field intensity (T) the Koenigsberger ratio, $Q = J_0/KT$ for most of these samples would exceed 0·5. Thus, according to the criterion proposed by Stacy (1967), the remanent magnetization of stable Dufek samples is dominated by grains of single-domain dimensions. Unstable samples from the same body have a similar range in J_0. K, and Q_n. Magnetic intensities in the Dufek intrusion are discussed in greater detail elsewhere (Beck & Griffin 1971).

5. Palaeomagnetism

Estimates based on rate of conductive heat loss indicate that the Dufek intrusion acquired its magnetization over a period of about 0·25 My. This should insure that non-axial elements of the geomagnetic field has sufficient time to average themselves out. The average direction of stable samples from the Dufek intrusion thus should record a fossil orientation of the axial dipole (Irving 1964, pp. 243–244). The 98 samples shown in Fig. 3 are magnetically stable, as shown by their response to a.c. demagnetization and the divergence of their mean direction from the present direction of the geomagnetic field. There appears to be no reason to suspect that their remanent

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magnetization is anything other than primary TRM, acquired upon original cooling. TCRM, a remanent magnetization acquired by mineralogical changes below the Curie temperature (Wasilewski 1969), also may be present, but if acquired on initial cooling would have virtually the same direction as conventional TRM. In view of these considerations, a palaeomagnetic pole position calculated from the mean of stable samples from the Dufek intrusion should indicate the position of the geographic pole with respect to East Antarctica roughly 168 ± 5 My ago, when the Dufek intrusion was formed.

The average direction for 91 stable *N* and *R* samples from the Dufek intrusion, after magnetic cleaning and tectonic correction, is shown in the first row in Table 2. Also shown are average directions for the same samples at four different demagnetization levels. Row 4 gives an alternative pole position calculation, based on the minimum dispersion criterion for selecting the optimum field for magnetic cleaning (Irving, Stott & Ward 1961). When the minimum dispersion criterion is applied to samples previously selected on the basis of the stable end-point method, the two methods not surprisingly give virtually identical results.

Table 2

Average direction of remanent magnetization and palaeomagnetic pole position for the Dufek intrusion

$\tilde{H}$	$\bar{D}$	$\bar{I}$	k	α_{95}	δ	λ'	ϕ'	m	p
SEP	53·0°	−69·0°	12·4	4·5°	23·0°	56·5° S	168·0° S	7·5°	6·5°
0	53·0	−67·5	10·3	5·0	25·5				
75	53·5	−68·5	11·3	4·5	24·0				
150	54·0	−69·0	11·8	4·5	23·5	56·5 S	167·0 W	7·5	6·5
300	49·0	−71·0	9·5	5·0	26·5				

Explanation: $\bar{D}$, $\bar{I}$, k, α_{95}, δ as described in Table 1. $\tilde{H}$, peak alternating field in a.c. magnetic cleaning; SEP indicates stable end point, described in Section 2; λ', ϕ' are latitude and longitude of palaeomagnetic pole; m, p are semi-axes of oval of 95 per cent confidence about the palaeomagnetic pole (Irving 1964, p. 69f). Tectonic correction as described in Table 1. *R* samples converted to equivalent *N* directions. Latitude and longitude of Dufek intrusion used in pole position calculation, 82·6° S, 50·0° W.

It was stated in the introduction that the weakest link in palaeomagnetic reconstructions of Gondwanaland is the apparent polar wandering curve for Antarctica. The present study does regrettably little to strengthen this link; the Dufek intrusion is Jurassic in age, and the Jurassic is to date by far the most thoroughly studied time unit in Antarctic palaeomagnetism. Knowledge of Jurassic palaeomagnetism to date has come from study of the Ferrar dolerites, which crop out in a broad belt extending some 4500 km along the Transantarctic Mountains (Stephenson 1966, map 4). Samples for palaeomagnetic studies have been collected at several localities within the belt. Results are summarized by Irving (1964, poles 9·37, 9·38, 9·42, 9·56). The mean of these pole lies at 139·5° W, 54° S, with $\alpha_{95} = 7{\cdot}5°$ and $k = 152{\cdot}7$ (Fisher 1953). A mean Jurassic pole for Antarctica may be computed by averaging the Dufek intrusion pole and the four Ferrar dolerite poles. This pole would lie at 145° W, 55° S, with $\alpha_{95} = 9{\cdot}0°$ and $k = 77{\cdot}2$.

Palaeomagnetic results discussed by McElhinny *et al.* (1968) and by McElhinny (1971) show that the pole was nearly stationary with respect to Africa from Early Triassic to Early Cretaceous time. Irving (1966) has summarized evidence for a hiatus in apparent polar wandering with respect to Australia in late Palaeozoic (Kiaman) and most of Mesozoic time. It would be interesting to investigate polar wandering relative to Antarctica for the same general time interval, but insufficient data are available for

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an adequate study. No poles are available for Antarctica for the late Palaeozoic. Early Triassic to middle Cretaceous time is represented by eight poles, if results from the Beacon Group sediments (Turnbull 1959; Bull, Irving & Willis 1962; Irving 1964, pole 9·39) are discarded on the grounds that they represent remagnetization by Ferrar dolerite intrusions. Several other difficulties also exist. For instance, it is unclear whether the four Ferrar poles should be treated as independent, inasmuch as the Ferrar dolerite may represent the same brief episode everywhere throughout the continent. Even more questionable is the ' pole ' for the Hawkes Rhyodacite Member of the Gambacorta Formation reported by Beck & Schmidt (1971). This ' pole ' is based on only four stable, independent samples, and its early Mesozoic age is inferred to be the result of reheating and remagnetization during a pre-Dufek orogeny (Ford 1971). The questionable Hawkes pole is shown in Fig. 7 for the sake of completeness; further sampling and laboratory study will be required to determine whether this is a valid Mesozoic palaeomagnetic pole. Finally, the Cretaceous poles reported by Scharon, Shimoya & Scharnberger (1969) and Scharnberger & Scharon (1971) are based on measurements of granite samples from West Antarctica. As discussed below, many authors (e.g. Schopf 1969; Hamilton 1967; Scharnberger & Scharon 1971) have suggested that West and East Antarctica were not part of the same continental block in pre-Tertiary time. If they were not, Mesozoic poles from West and East Antarctica cannot be compared without allowance for continental drift. In view of these difficulties, perhaps the most conservative estimate of the mean Mesozoic pole for East Antarctica is a point midway between the composite Ferrar dolerite pole and the Dufek intrusion pole. This point lies at 54° S, 153° W. The pole for the Dufek intrusion and the composite Ferrar dolerite pole are statistically distinct, indicating

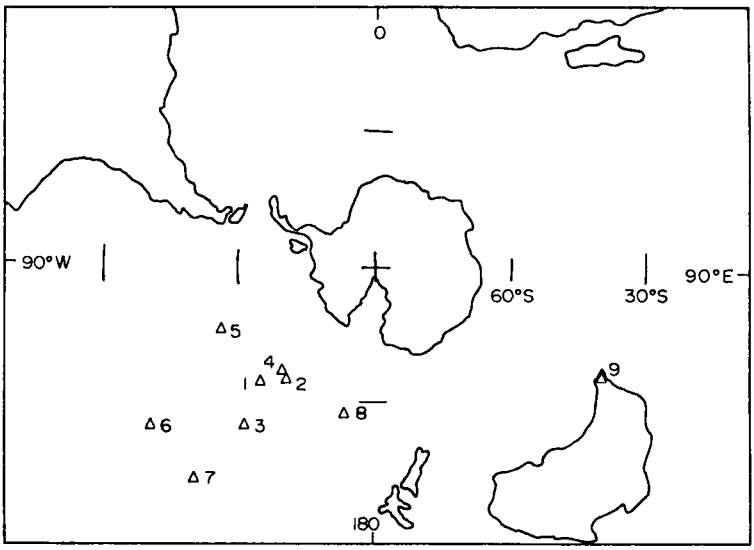


FIG. 7. Mesozoic palaeomagnetic poles for Antarctica. 1, Ferrar dolerite from the Theron Mountains and Whichaway Nunataks (Blundell & Stephenson 1959; Blundell 1966); 2, Ferrar dolerite from Ferrar Glacier (Turnbull 1959); 3, Ferrar dolerite from Wright and Victoria Dry Valleys (Bull & Irving 1960; Bull *et al.* 1962); 4, Ferrar dolerite from Beardmore Glacier (Briden & Oliver 1963); 5, Questionable early Mesozoic pole from the Hawkes Rhyodacite Member of the Gambacorta Formation, Pensacola Mountains (Beck & Schmidt 1971); 6, 7, Cretaceous granitic rocks from Ellsworth Land (Scharon *et al.* 1969); 8, Dufek intrusion (this study); 9, Upper Cretaceous rock from Marie Byrd Land (Scharnberger & Scharon 1971).

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that the gabbro and dolerite probably should not be ascribed to a single brief pulse of magmatic activity.

McElhinny (1971) calculated a mean of nine Mesozoic poles for Africa with a precision parameter (Fisher 1953) of 111. The same statistic for Australian Mesozoic pole (numbers 10–18 from Table 3 of McElhinny 1971) is 73. The nine Antarctic Mesozoic poles shown in Fig. 7 are widely scattered, with a precision of only 9. Most of this scatter is introduced by the Cretaceous poles from West Antarctica, which are considerably outside the cluster of East Antarctic poles. This observation tends to support the suggestion that there has been movement of West Antarctica relative to East Antarctica since the Cretaceous. However, the pole (Blundell 1961) for the Andean intrusive suite from the Antarctic peninsula in West Antarctica is not significantly different from the present pole. The age cited for the Andean intrusive suite is Late Cretaceous to Eocene. Halpern (1962) has obtained K–Ar dates of 75 ± 8 and 100 ± 20 My for Andean intrusive rocks. These observations suggest that there has been no significant movement of the Antarctic Peninsula relative to the pole in at least the last 75 My. Data from Marie Byrd Land, on the other hand, seem to require a lengthy polar transition from the Cretaceous until the present. Thus, taken at face value, palaeomagnetic results for the Mesozoic and Tertiary of Antarctic support the proposition that East and West Antarctica have had different histories with respect to continental drift until fairly recent times, and also suggest that there has been some degree of internal deformation within West Antarctica.

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